

Risk Prediction Models: Temporal Recalibration and Non-Linear Effects

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Abstract

Temporal recalibration updates an existing risk prediction model to allow for changes in underlying risk levels. It relies on the assumption that the effect of the risk factors does not change with time. A predictive model is created using data from the full time-period to get a precise estimate of the effect of the risk factors. The model is updated to reflect the survival probabilities seen in the latter period, to more accurately predict outcomes from this point on. There is also a focus on identifying and incorporating non-linear effects associated with certain variables in the model. Failure to incorporate non-linear effects can lead to time-dependent effects when the risk factor is non-linear. Accurate modelling of the risk factors and survival outcomes of diseases like cancer are key for prevention and to guide treatment decisions. This project analyses the breast cancer data set from the Rotterdam study (Royston and Altman (2013)) [1] that looked at outcomes following surgery, for surgeries performed between 1978 and 1993. The data set is split into two time periods based on the year in which surgery was performed, 1978-1987 and 1988-1993. The two periods are examined to determine if survival probabilities are the same in both periods. Results show that survival outcomes are improved in the later time period when compared to the earlier one.

The associated risk factors are assessed, and it is considered how best to model the impact that these risk factors have over time. The effect of age at surgery is shown to be non-linear. Increasing age improves survival up to middle age and significantly disimproves survival probabilities as age gets older than that. The effect of age also varies over follow-up time. As a results age was included as a non-linear term in the prediction model.

Temporal recalibration was shown to be successful in updating the risk prediction model, developed on a long history of data, where more recent patients in the data set had improved survival outcomes compared to the earlier patients. Temporal recalibration can be applied when there are competing risks. Further study is needed to apply temporal recalibration to Markov processes.

1. Introduction

My project is centred around risk prediction models and temporal recalibration. The risk prediction models utilised are based on data collected in the Rotterdam breast cancer study and developed using time-to-event (survival) analysis, where the event in question is death or relapse. These models provide a survival prognosis for an individual, based on certain risk factors such as age and cancer grade, and give an understanding of how the different risk factors impact survival.

Recalibration refers to adjusting a given model produced in a certain setting so that it can be applied in another setting. This report looks at temporal recalibration which involves updating an older model so it can be applicable currently. Temporal recalibration is only necessary if survival outcomes are changing over time. In this paper, I will compare the two time periods in the data set, analysing survival outcomes and the impact of certain risk factors in each period. Temporal recalibration assumes that although the underlying risk may change, the relative effect of the risk factors remains the same. I will investigate whether this is the case using statistical techniques that identify time-dependent effects that violate the assumption of proportional hazards. I will present an in-depth analysis of the association between risk factors and risk that identifies non-linear associations. I will highlight the equivalence between non-linear covariate effects and time-dependent effects. Then, temporal recalibration will be performed on the full risk prediction model to more closely reflect the survival outcomes seen in the latter time period.

In chapter 2, I will give an overview of the data available and a breakdown of the distribution of the variables used. In chapter 3, I will describe the statistical methods used in this report. The results will be presented in chapter 4. In chapter 5, the results will be discussed, and potential future areas of research will be alluded to.

2. Data

The specific data set used here comes from a paper by Royston and Altman (2013) [1] that looked at the external validation of a Cox prognostic model. It follows 2982 primary breast cancer patients who had surgery between 1978 and 1993. The variables in this data set are a patient identifier, year of surgery, age at surgery, menopausal status (premenopausal, postmenopausal), tumour size ($\leq 20\text{mm}$, $20\text{-}50\text{mm}$, $>50\text{mm}$), grade, number of positive lymph nodes, progesterone receptors, oestrogen receptors, hormonal treatment (no, yes), chemotherapy (no, yes) days to relapse or last follow-up, relapse (no, yes), days to death or last follow-up and death (alive, dead). Figure 1 describes the distribution of the continuous variables age, number of positive lymph nodes, oestrogen receptors and progesterone receptors. The distribution of age is relatively symmetrical, while the other three risk factors are considerably skewed.

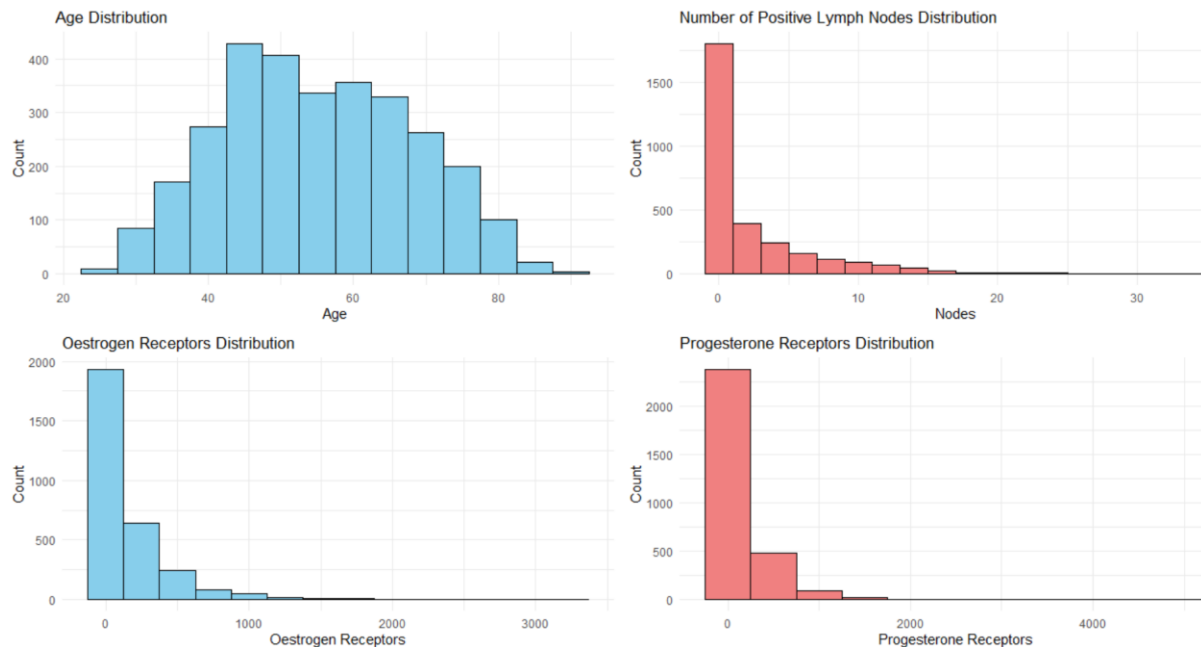


Figure 1: Histograms illustrating the distribution of continuous risk factors in the Rotterdam study

See below table showing averages and deviations of variables across each period.

Table 1: Descriptive statistics (mean(sd), median (Q1, Q3) or number(percentage)) for Rotterdam study 1978-1987 and 1988-1993

Variable name	Definition	Period 1 (1978-1987)	Period 2 (1988-1993)
Age	Years	56.3 (13.1)	54.3 (12.8)
Progesterone receptors	fmol/l	44 (8, 184)	40 (3, 206)
Oestrogen receptors	fmol/l	58 (12, 195)	63 (11, 206)
Positive lymph nodes	Number	1 (0, 4)	0 (0, 3.5)
Tumour size	<= 20mm	459 (39.3%)	928 (51.1%)
	20-50mm	567 (48.6%)	724 (39.9%)
	> 50mm	141 (12.1%)	163 (9%)
Tumour grade	grade 2	344 (29.5%)	450 (24.8%)
	grade 3	823 (70.5%)	1365 (75.2%)
Menopausal status	premenopausal	467 (40%)	845 (46.6%)
	postmenopausal	700 (60%)	970 (53.4%)
Hormone treatment	no	1120 (96%)	1523 (83.9%)
	yes	47 (4%)	292 (16.1%)
Chemotherapy	no	927 (79.4%)	1475 (81.3%)
	yes	240 (20.6%)	340 (18.7%)

3. Methods

In this chapter, I will describe the methods used in this report. Guidance was taken from lecture notes for ST3054 Survival Analysis with Dr. Rabia Naqvi in describing certain sections of the following chapter, namely 3.1 Cox Proportional Hazards Model and 3.2 Kaplan-Meier Function.

3.1 Cox Proportional Hazards Model

The Cox Proportional Hazards model is used to assess the probability of survival or death within a certain timespan and the effect of certain explanatory variables (e.g. age, gender, smoking status etc...) on survival probabilities. The Cox model defines the hazard function for the i^{th} individual as:

$$h(t; z_i) = h_0(t) \exp(\beta z_i^T)$$

where:

$h_0(t)$ is the baseline hazard i.e. the hazard for the baseline individual

- z_i refers to the risk factors present for the i -th life, and
- β denotes the coefficients associated with those risk factors

Here, the hazard function represents the instantaneous risk of an event e.g. death, given that the individual has survived up to time t . Mathematically, this can be written as:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

This represents the probability that the event occurs as time T which is within the interval between t and $t + \Delta t$ with the limit of Δt approaching 0.

The coefficient β represents the logarithm of the hazard ratio associated with a one-unit increase in the corresponding covariate z_i . If this coefficient is positive, then the hazard increases with this risk factor and vice versa. To estimate β , we maximise the likelihood function. Let $R(t_j)$ denote the set of lives at risk just before the j -th observed lifetime. This likelihood function can be written as:

$$L(\beta) = \prod_{j=1}^k \frac{h_0(t_j) \exp(\beta z_j^T)}{\sum_{i \in R(t_j)} h_0(t_j) \exp(\beta z_i^T)}$$

Here, the baseline hazard $h_0(t_j)$ cancels on the top and bottom, leaving:

$$L(\beta) = \prod_{j=1}^k \frac{\exp(\beta z_j^T)}{\sum_{i \in R(t_j)} \exp(\beta z_i^T)}$$

This is described as the partial likelihood because it calculates the likelihood that an event occurs to a particular person, given that an event has occurred to someone at this time. However, this does not account for the baseline hazard or the cumulative survival probability.

3.2 Kaplan-Meier Function

Parametric and non-parametric methods can be used to estimate the hazard and cumulative survival.

One non-parametric method is using the Kaplan-Meier model. Let d_j be the number of deaths observed at failure time t_j , and we define the hazard function by:

$$h_j = P(T = t_j \mid T \geq t_j)$$

This hazard can be estimated within the interval containing failure time t_j as:

$$\hat{h}_j = \frac{d_j}{n_j}$$

This is the Maximum Likelihood Estimate (MLE) of the likelihood function:

$$L(\hat{h}) = \prod_{j=1}^k (h_j^{d_j} (1 - h_j)^{n_j - d_j})$$

Thus, the estimated survival function, which is the summation of one minus the hazard up to a certain point (time t) i.e. the probability of surviving at least until t is:

$$\hat{S}(t) = \prod_{t_j \leq t} \left(\frac{n_j - d_j}{n_j} \right)$$

3.3 Cubic Splines

Spline models can also be used to model the baseline hazard in flexible parametric survival models [2]. We choose a set of knots to divide the time axis into intervals, then create a spline basis $B_1(t), B_2(t), \dots, B_k(t)$ for $h_0(t)$. Each basis function corresponds to a segment of the time axis and is defined by a cubic polynomial. We write $h_0(t)$ as a weighted sum of the spline basis functions:

$$h_0(t) = \exp \left(a_0 + \sum_{j=1}^k a_j B_j(t) \right)$$

where:

- a_0 is an intercept term,
- a_j are the coefficients for the spline basis functions,
- $B_j(t)$ are the spline basis functions

Cubic splines are a common tool used in survival analysis, that can also be used to model non-linear associations between risk factors and survival. A cubic spline is a series of cubic polynomials that join smoothly at specified points known as knots. The spline basis functions $B_1(x), B_2(x), \dots, B_k(x)$ are constructed over these knots, and the hazard ratio of risk factor is expressed as a weighted sum of these basis functions:

$$HR(x) = \exp \left(\sum_{j=1}^k \beta_j B_j(x) \right)$$

where the β_j coefficients are estimated by the model. This allows the hazard ratio to be modelled flexibly without forcing linearity.

3.4 Fractional Polynomials

Fractional polynomials are used to model non-linear relationships in variables. In the context of this report this could relate to parametric modelling of the baseline hazard function or alternatively non-linear associations between the hazard and the risk factors. To identify and model non-linear risk factors, the multiple fractional polynomial (`mfp()`) command iterates through each defined variable exploring various fractional polynomial transformations to find the best fitting form. It compares a linear representation of the variable to the model without it and tests if the addition of a linear representation of the predictor statistically significantly improves the model by reducing deviance. Similarly, it will compare fitting the model with a power term x^p where the following powers are considered -2, -1, -0.5, 0, 0.5, 1, 2, 3. Also, it will compare a model with two power terms to the model with one. In the case of a Cox model, the partial likelihood will be used to test the statistical significance of a reduction of deviance in the model, and the simplest suitable one will be chosen, e.g. the model with two power terms will only be chosen if it significantly outperforms the other three simpler models.

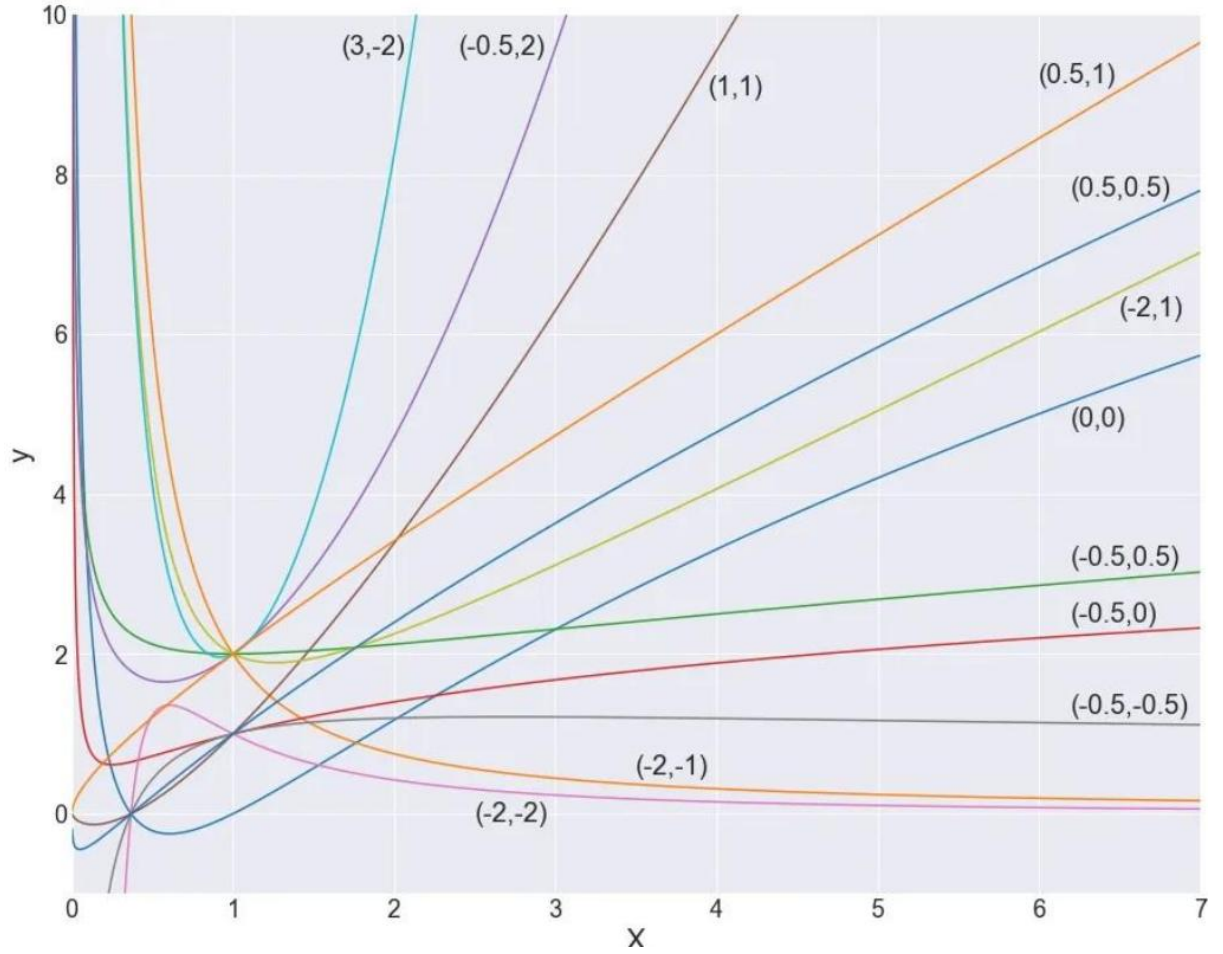


Figure 2: Degree 2 fractional polynomials with all β -coefficients = 1 for various power terms. [3]

Fractional polynomials can also be used to model the baseline hazard function in flexible parametric models [2]. A set of fractional powers are chosen, which will serve as the basis for modelling $h_0(t)$. The baseline hazard $h_0(t)$ is defined using a combination of these terms:

$$h_0(t) = \exp \left(a_0 + \sum_{j=1}^p a_j t^{p_j} \right)$$

where:

- a_0 is an intercept term,
- a_j are the coefficients for the fractional polynomial terms,
- p_j are the chosen powers, which can be fractional (-2, -1, -0.5, 0, 0.5, 1, 2, 3).

3.5 Schoenfeld Residuals

The proportional hazards assumption is used in the Cox proportional hazards model. This means that the hazard ratio for each variable is assumed to be constant over time. Consider a patient A with a grade 2 tumour and a patient B, all of whose risk factors are identical to patient (same age etc...) but with a grade 3 tumour, if the hazard ratio for grade is 1.5, then patient B is 1.5x more likely to have an event. The proportional hazards assumption dictates this ratio of the hazard of patient B relative to patient must always be 1.5 and this does not change over time.

Schoenfeld residuals measure the difference between the observed and expected values of a covariate at the time of an event. These are used to measure if the effect of a covariate is changing over time, or if the Cox model, which constrains the risk factors to be constant over time, is accurately representing the hazard ratios at each given time. If the proportional hazards assumption holds, Schoenfeld residuals should be randomly scattered around zero. In my analysis a loess smoothening was added to my plot of Schoenfeld residuals against time, where a flat line would support the proportional hazards assumption. However, a clear upward or downward trend in the plot would suggest the hazard ratio changes with time. Similarly, the Schoenfeld residuals can be tested statistically to produce a p-value, representing how likely or unlikely the residuals we get would be to occur if hazard ratios were constant. To test at 95% confidence, a p-value < 0.05 would lead to a rejection of the proportional hazards assumption.

3.6 Temporal Recalibration

Temporal Recalibration is a technique that can be useful in scenarios where survival outcomes are changing over time, but the relative effect of risk factors remains constant [2]. Predictive models are usually created on data sets spanning long periods of time and these models can become out of date. Generally, this occurs in areas where survival is improving over time, leading to existing models to underestimate survival probabilities. Cox proportional hazard models, of the form

$$h(t; x_i) = h_0(t)e^{\beta x_i}$$

are often used for this predictive modelling, where the left side of the equation represents the hazard function and the right side the baseline hazard function multiplied by the exponential of the prognostic index. Flexible parametric survival models can also be used to produce smooth curves where proportional hazards is not required meaning time-dependent predictor effects can be included and the baseline hazard can be modelled parametrically for example using splines. Such a model can be represented as

$$\ln H(t; x_i) = f(\ln t | c, k_0) + \beta x_i$$

where the log of the cumulative hazard function is equal to the restricted cubic spline function of the baseline hazard + the prognostic index. Period analysis describes only using a more recent subset of the data to form the predictive model to make it more up to date. This comes with the drawback of a reduction in sample size and so there is a trade-off to be made with the size of the window selected, in that the more up to date the model, the smaller the sample size and wider the confidence intervals. Temporal recalibration is a method that can be used to combine the greater sample size of the full cohort analysis with the more up to date prediction of the period analysis. First a model is developed using the full cohort dataset to get an estimate of the predictor effects. Then, the model is recalibrated by re-estimating the baseline hazard using the more recent data from the period analysis while holding the predictor effects previously estimated fixed. In the case of a flexible parametric survival model this can be written as

$$\ln H_{new}(t; x_i) = f_{new}(\ln t | c, k_0) + \text{offset}(PI_i)$$

where $\text{offset}(PI_i)$ is the predictor effects previously estimated kept fixed by defining it as an offset term. The performance of such models can be assessed, where a full cohort model, period analysis model and temporal recalibration model can be implemented at a given time, and expected outcomes of each model can be compared to observed outcomes over the following years. Crucially, the outcomes being tracked would be outcomes of patients diagnosed at the time of the forming of the models. Through such a study, temporal recalibration can be shown to provide more accurate survival predictions for newly diagnosed patients than full cohort analysis.

3.7 Steyerberg's Steps for Model Development

Inspiration for the structure of the methodologies carried out was taken from Ewout W. Steyerberg's seven steps to model development (Steyerberg and Vergouwe (2014)) [4].

These steps were designed with a particular focus on logistic regression, which is used to predict binary outcomes, such as 30-day mortality. In logistic regression, the probability of an event can be modelled as: $\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$. Here, β_0 is the intercept, and the β_j are the coefficients for the predictors X_j .

However, these steps can be applied similarly to time-to-event data. The temporal recalibration that will be performed using Cox regression, where the baseline hazard $h_0(t)$ is updated, is analogous to changing the β_0 intercept term in a logistic regression model

Step 1: Problem definition and data inspection

The objective was to model breast cancer survival and investigate temporal changes in prognostic effects and overall survival, using the Rotterdam data set, to determine if temporal recalibration was applicable. The distribution of each risk factor was explored across the two periods and survival curves were drawn for each period.

Step 2: Coding of predictors

All the continuous risk factors were modelled linearly to begin with, before age and grade were focused on. The importance of age was scaled per decade for interpretability, rather than reporting hazard ratios for a one-year increase. Fractional polynomials and cubic spline representations were explored for age and a non-linear modelling was shown to be most accurate, with a fractional polynomial used. Size was inputted as a categorical variable with 3 categories, where <20mm was the reference, to which the other two size categories were compared.

Step 3: Model specification

Interaction terms and potential time-dependence were investigated, and the proportional hazards assumption was tested. Age was split into 4 groups to visualise survival curves for each separately. Schoenfeld residuals were used to provide a hazard

ratio plot over time, and to test the proportional hazards assumption statistically. With age being shown to have a non-linear effect, the multiple fractional polynomials command was used to fit a Cox proportional hazards model, using deviance minimisation to select fractional polynomial degrees for age, choosing linear representations for some other variables, and choosing to disregard variables not deemed to have a significant impact on survival.

Step 4: Model estimation

Model estimation was performed by maximising the partial likelihood, to acquire β coefficients for the terms specified in the model. Previous to estimating the full model with linear and non-linear representations, β coefficients were estimated for a naïve model with all terms included linearly, models with just age and grade as risk factors, separate models for each period, a model with period included as a risk factor, models with interaction and time-dependent terms and models based on the data for each grade of tumour. Each of these models provided important information on the data and how to construct a viable prediction model.

Step 5: Model performance

The performance of the temporally recalibrated model, in achieving its goal of updating the full cohort model to reflect improved survival outcomes, is shown by plotting the cumulative incidence curve (the inverse of the survival curve) of the full cohort model, the recalibrated model and the period analysis model. Through this the survival outcomes in the temporally recalibrated model are shown to more accurately reflect the outcomes seen in the period analysis model based off the data in the later period.

Step 6: Model validity

The validity of the temporally recalibrated model was tested by applying the prognostic index calculated in the full cohort model, as a risk factor in a Cox model based just on the latter period. This is done as an assessment of model validity that Steyerberg refers to as calibration-in-the-large. A coefficient near one for the prognostic index confirmed good calibration-in-the-large and supported the rationale that the estimated risk factors

coefficients from the full data set could be used in the temporally recalibrated model, and that just changing the baseline hazard was valid.

Step 7: Model presentation

Results of the temporal recalibration were shown in the cumulative incidence curve of the three final models. Estimates of hazard ratios were presented throughout in tables and plotted temporally using cubic splines and fractional polynomials.

Though not performed in this project, a reasonable way to present a model predicting survival probabilities over time according to a set of risk factors would be a web-based calculator.

4. Results

We consider a temporal recalibration of the well-known Rotterdam Study [5]. To the best of my knowledge, this is the first time this has been done. The Rotterdam study is a prospective cohort study aimed at understanding the causes and risk factors of chronic diseases such as cardiovascular and neurological diseases as well as cancer. This study was initiated due to the presence of increasingly aging populations which would lead to an increase in the prevalence of these diseases. Participants undergo baseline as well as follow-up assessments to track disease progression and help researchers understand the key risk factors associated with morbidity. The study provides detailed and long-term follow-up data making it very useful for survival analysis.

4.1 Risk Factors

The distribution of the risk factor is described in section 2. A Cox model produced with all the variables included does not show a statistically significant impact on survival for the variables associated with chemotherapy, hormonal treatment and oestrogen receptors, or menopausal status. The remaining variables do appear to have a statistically significant effect. The following analysis looks at the data over the full time period, 1978-1993.

Table 2: Hazard ratios for each variable

Risk factor	Hazard ratio	95% CI	p-value
Age ¹ (years)	1.15	1.07, 1.24	<0.001
Grade (3 vs 2)	1.37	1.19, 1.58	< 0.001
Size (20-50mm) ²	1.56	1.37, 1.77	< 0.001
Size (>50mm)	2.28	1.90, 2.72	< 0.001
Nodes	1.08	1.07, 1.09	< 0.001
Progesterone receptors (fmol/l)	0.9996	0.9994, 0.9999	0.0027
Oestrogen receptors (fmol/l)	0.9999	0.9997, 1.0002	0.62
Hormone treatment	0.94	0.79, 1.11	0.46
Chemotherapy	1.05	0.90, 1.23	0.54
Menopausal status (pre, post)	1.07	0.88, 1.31	0.48

¹ Hazard ratio associated with a 10-year increase in age

² Both size hazard ratios are relative to a tumour of size <20mm

Age is adjusted by subtracting 50 from age and then dividing by 10. Subtracting 50 equates a 50-year-old to 0, so that a 50-year-old can be used as the baseline, this does not affect the hazard ratio which will still have the same value for a one-unit increase. Dividing by 10 does affect the hazard ratio, it makes a one unit increase in the variable equivalent to a 10-year increase in age, so the hazard ratio shows the effect of a 10-year increase in age. This can be done as a 1-year increase in age generally doesn't have drastic impact.

4.2 Period Comparison

A Kaplan-Meier curve was drawn for each period, comparing survival for each, ignoring risk factors, see figure 3. Confidence intervals were included, showing improved survival outcomes for those in the later period. The p -value < 0.0001 shows demonstrable difference between the two periods.

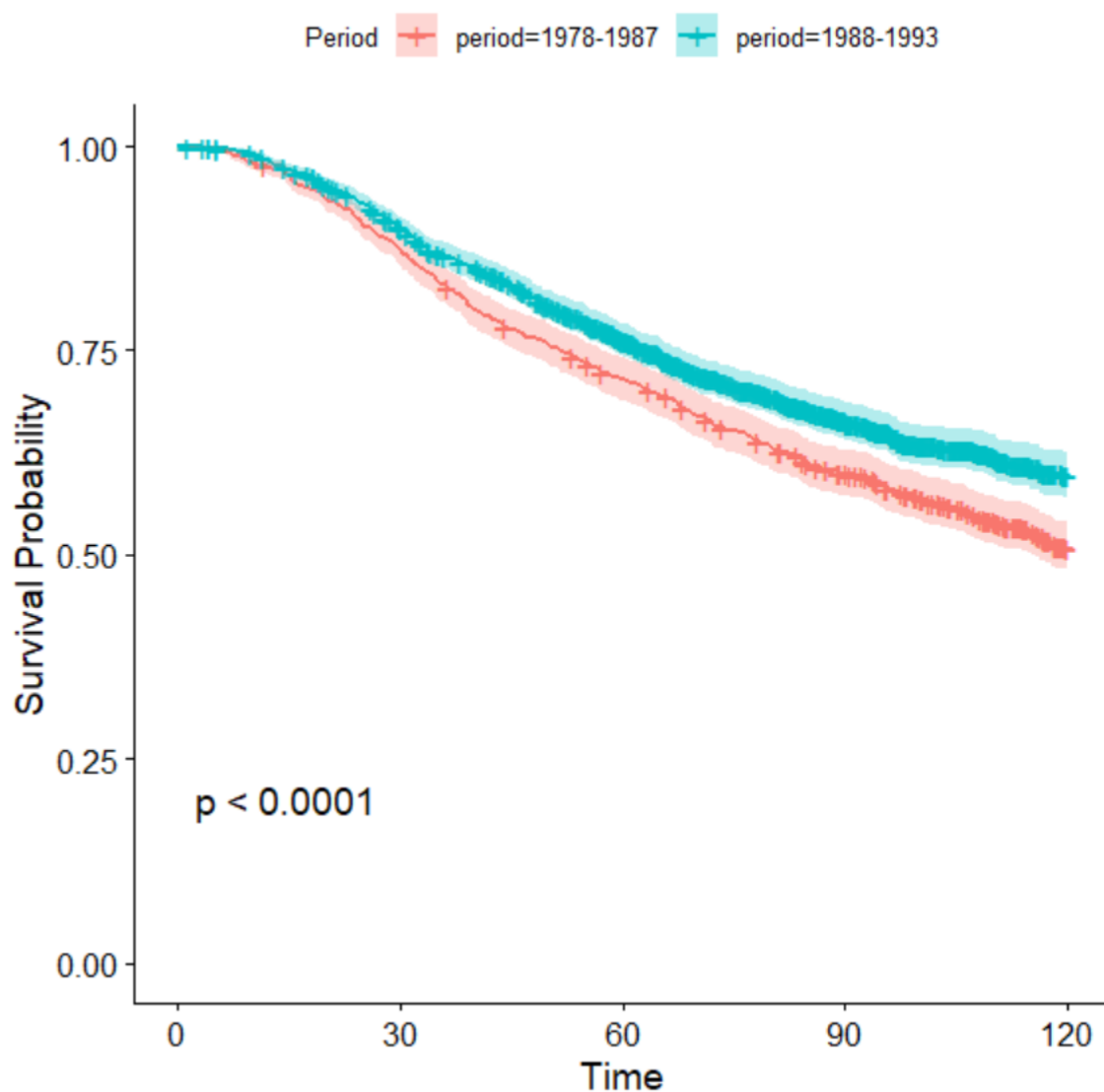


Figure 3: Kaplan-Meier survival curves for the periods 1978-1987 and 1988-1993

Motivated by these results we considered the distribution of risk factors in the two time-periods. A small difference was discovered in the average age across the two periods, with 56.3 the average age with a standard deviation of 13.1 in the first period 1978-1987, and 54.3 with a standard deviation 12.8 the average in the second period 1988-1993.

This small difference in age could be thought to explain some of the difference in the Kaplan-Meier curves, with the slightly older first period having worse outcomes. When we look at the breakdown of the grade of tumours, we see a smaller proportion of grade 3 tumours in the first period relative to the second (70.5% vs 75.2%). This would suggest a marginal improvement in survival in the latter period.

Similar Kaplan-Meier curves were produced for the two grades showing clear difference in survival outcomes for participants with either grade, with much better outcomes for those with grade 2.

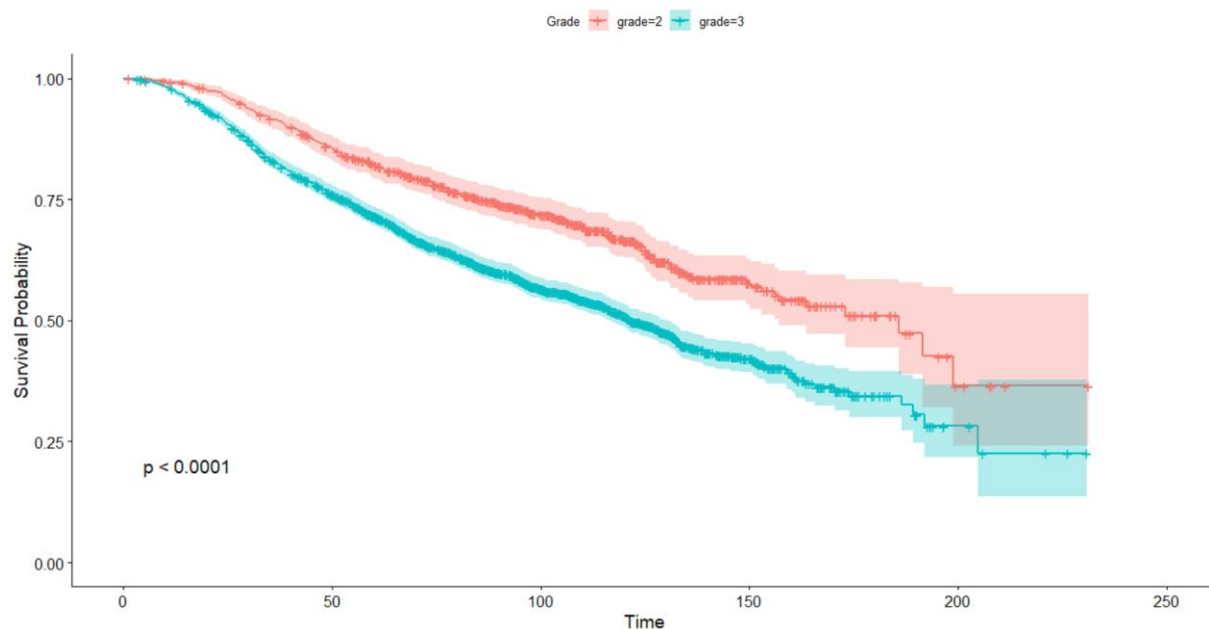


Figure 4: Kaplan-Meier survival curves for tumours of grade 2 and grade 3

To focus on modelling of time-dependent and non-linear effects, the following section will focus on grade and age.

A Cox model was produced using the full data set across both periods, taking age and grade as risk factors. The hazard ratios for this model are shown below

Table 3: Hazard ratios for model including age and grade in full dataset.

Risk factor	Hazard ratio	Confidence interval
Age ¹	1.20	1.15, 1.25
Grade	1.63	1.42, 1.87

We see death is 1.63 (1.42, 1.87) times as likely for a person of the same age with a grade 3 tumour relative to a grade 2 tumour. We can also run this model including time period as a risk factor.

To examine the effect of period, a model was then produced with period included as a risk factor. Table 4 shows the hazard ratios for this model

Table 4: Hazard ratios for Cox model with period included as a risk factor

Risk factor	Hazard ratio	Confidence interval
Age ¹	1.19	1.14, 1.24
Grade	1.65	1.44, 1.89
Period	0.81	0.72, 0.90

Here the hazard ratio is 0.81 (0.72, 0.9) for the period 1988-1993, relative to the period 1978-1987. This shows demonstrable improvements in survival for those in the later period. The inclusion of period as a covariate did not substantially alter the hazard ratios for age and grade, indicating the robustness of these predictors across time. The consistency of these hazard ratios across the two models suggests that the improvements in survival outcomes seen are not driven by changes in the prognostic relevance of these variables, but likely by other clinical or treatment-related factors.

Then, separate Cox models were produced for each period, which can show how the impact of the risk factors varies across the two periods. In doing this, we discover grade appears to be a stronger predictor in the second period than the first, and age stronger in the first. This would mean that the difference in outcomes between grade 2 and grade 3

participants is larger in the second period than the first. The hazard ratio of a grade 3 tumour relative to a grade 2 tumour in the first period is 1.43 (1.19,1.7), with the same figure in the second period being 1.97 (1.59,2.44).

Table 5: Hazard ratios for Cox models produced on the data from each time period

Risk Factor	Period	Hazard ratio	Confidence interval
Age ¹	Period 1 (1978-87)	1.26	1.18, 1.34
	Period 2 (1988-93)	1.12	1.06, 1.20
Grade	Period 1 (1978-87)	1.43	1.19, 1.7
	Period 2 (1988-93)	1.97	1.59, 2.44

The baseline survival curve was looked at for both periods. This shows how the survival probability relates to time for a 50-year-old with a grade 2 tumour.

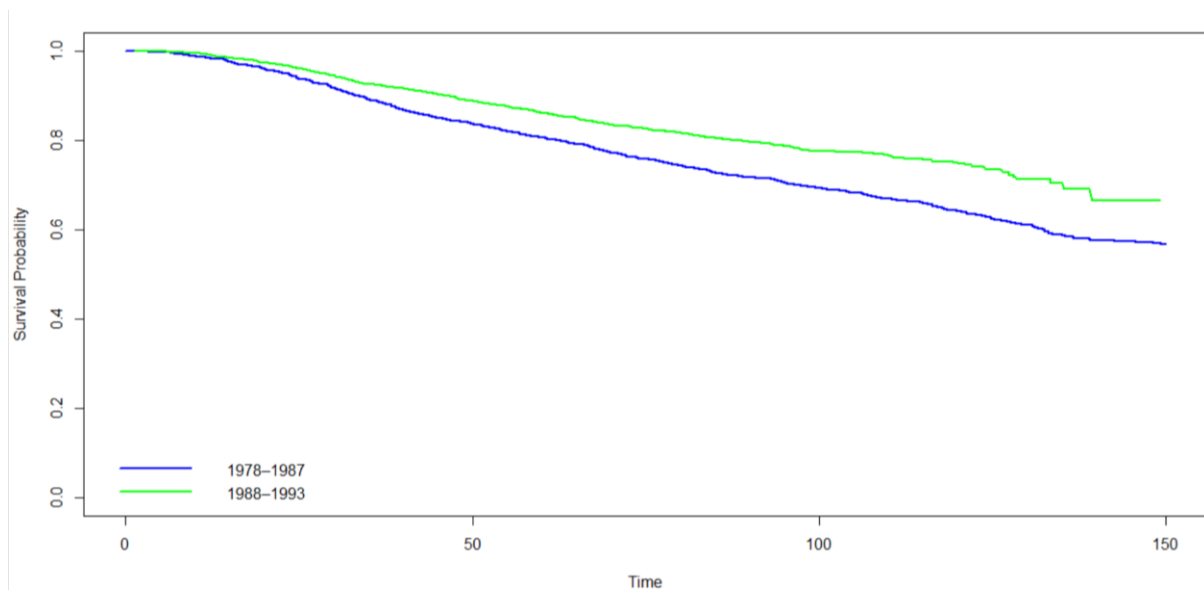


Figure 5: Baseline survival curves for each period

This appears to show again improved survival probabilities in the latter period when compared to the first.

4.3 Hazard Ratio Modelling Over Time

In the Cox regression models, it was presumed that hazards are proportional over time. When tested statistically using the Schoenfeld residuals, a very small p-value of 0.001 for age shows that this risk factor violates the proportional hazards assumption, meaning the effect of age on hazard changes over time. The p-value for grade is borderline (.057) when testing at 95% confidence, suggesting it may violate the proportional hazards assumption.

The proportional hazards assumption can also be tested graphically using $\log(-\log(S(t)))$ curves. Under the PH assumption, the two curves should remain parallel. Relatively parallel lines are seen when doing this for grade.

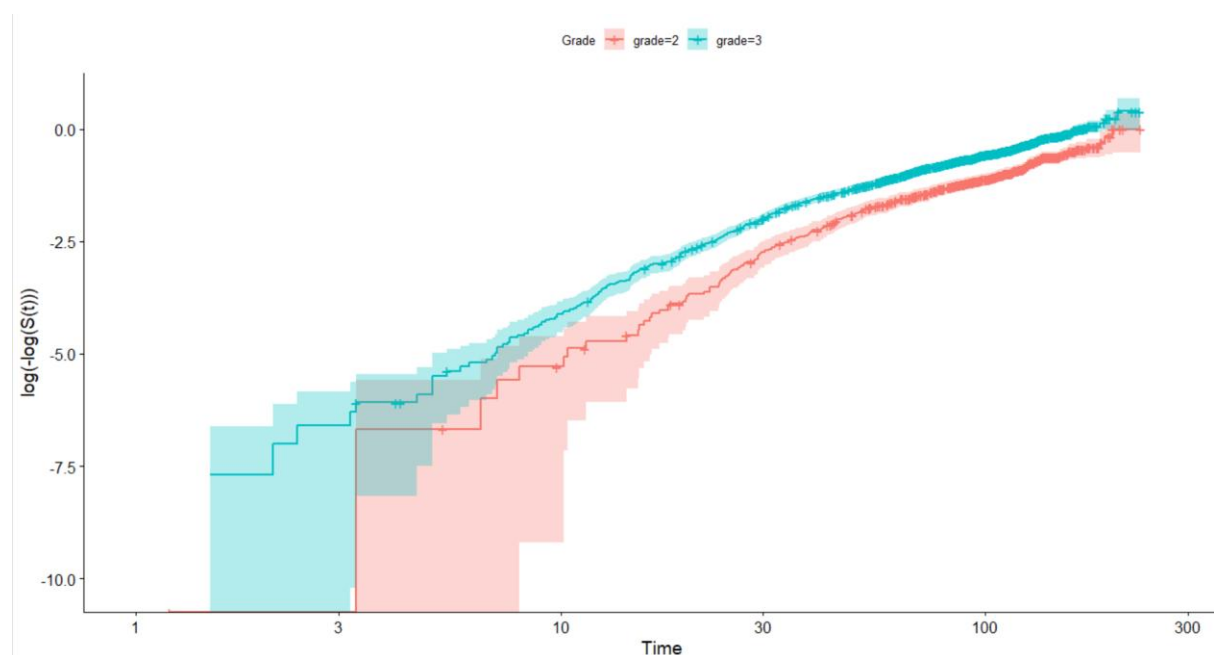


Figure 6: $\log(-\log(S(t)))$ curves for each grade

To assess how hazard ratios are changing over time, scaled Schoenfeld residuals were plotted over time, with a loess smoothing effect added for visualisation. A straight line would back up the proportional hazards assumption, whereas as a fluctuating line would show that hazard ratios are not constant. The plot of grade over time shows possible early fluctuation before a relatively smooth straight line emerges.

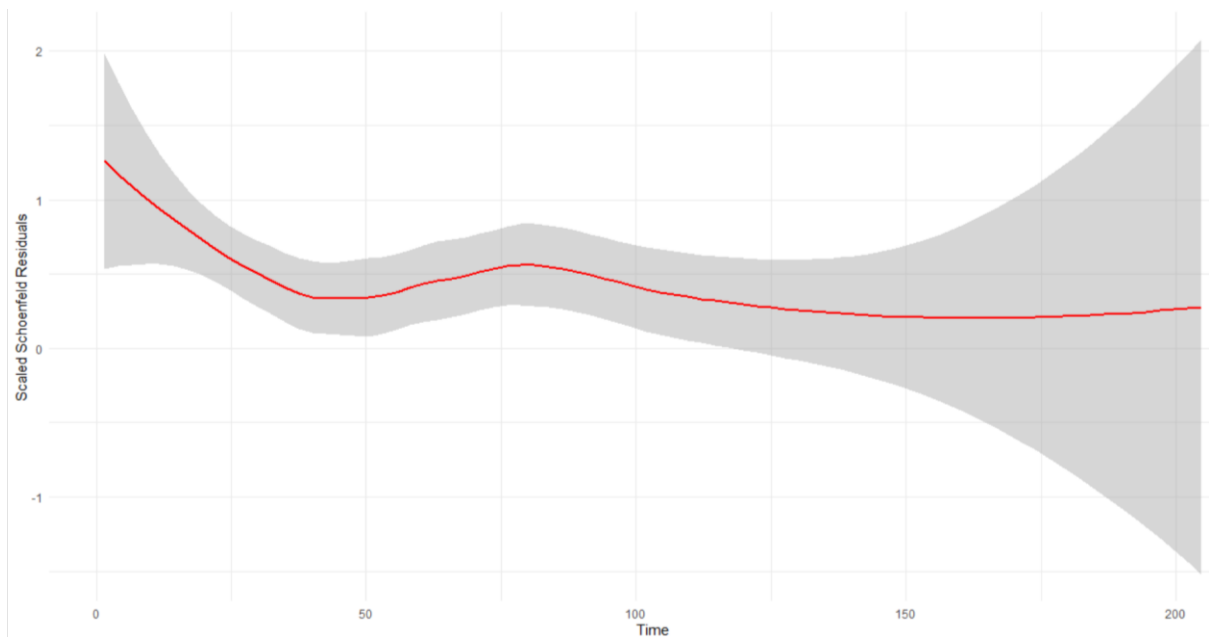


Figure 7: Scaled Schoenfeld residuals of grade over time

When age is split into quartiles, we obtain the curves in figure 8 against $\log(\text{time})$. Under the PH assumption, the curves of these four groups should remain parallel.

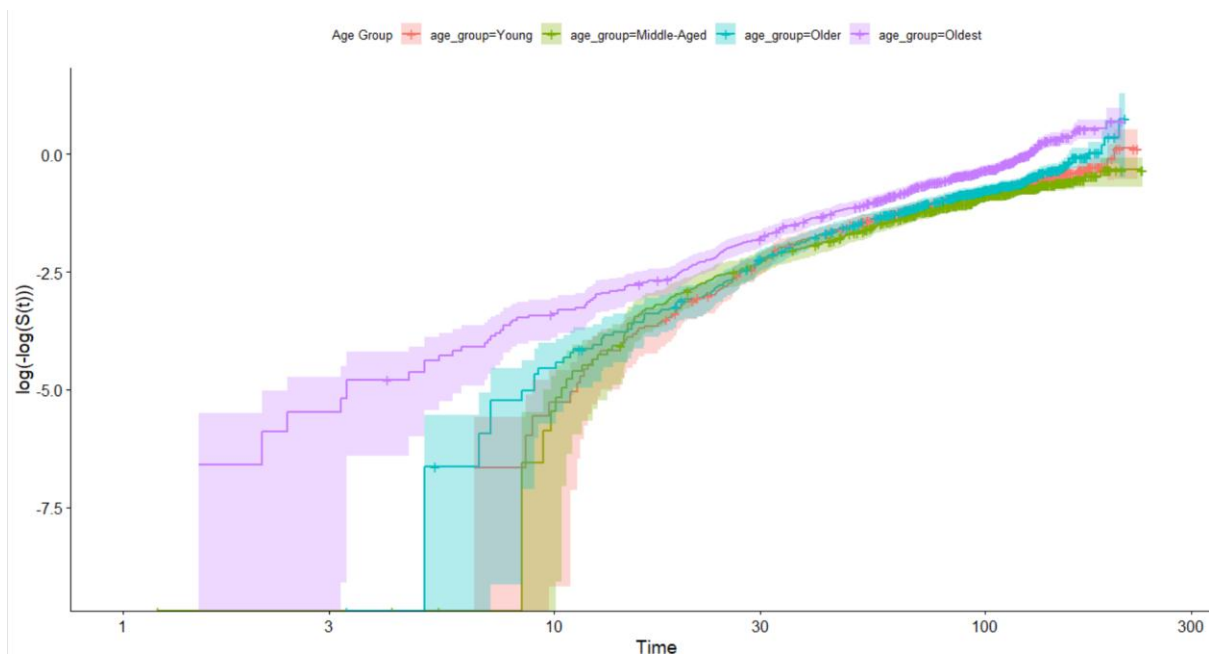


Figure 8: $\log(-\log(S(t)))$ curves for each age group

While there is some overlap early on, there are also large confidence intervals. When plotting the scaled Schoenfeld residuals for age, the curve appears to be dropping initially, before rising with time. This confirms age violates the proportional hazards assumption.

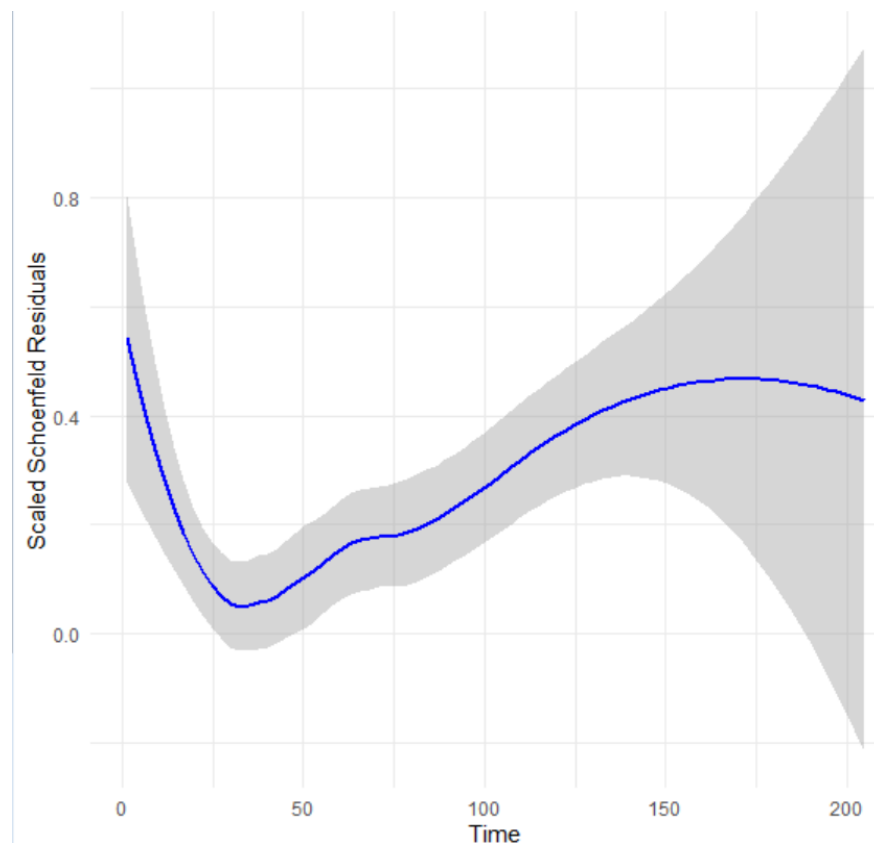


Figure 9: Scaled Schoenfeld residuals of age over time

4.4 Interaction Effects

To investigate if there is interaction between age and grade, first two Cox models were created, with grade equal to 2 for all data in the first, and 3 in the second, and the effect of age in each was compared. The hazard ratio associated with age in each model is presented in table 6.

Table 6: Hazard ratio for age based off two models of the data with two different grade values.

Risk Factor	Grade	Hazard ratio	Confidence interval
Age ¹	Grade 2	1.028	1.018, 1.039
	Grade 3	1.012	1.007, 1.017

The effect of age appears to be larger in the first model with the grade 2 cohort.

A Cox model was then created with an interaction term between age and grade. Hazard ratios and p-values for each term are described in the table below.

Table 7: Hazard ratios for Cox model with age and grade interaction term

Risk factor	Hazard ratio	Confidence interval	p-value
Age ¹	1.36	1.23, 1.5	<0.001
Grade	1.82	1.55, 2.14	<0.001
Age ¹ : Grade	0.86	.77, .96	0.006

The p-value of 0.006 for the interaction term confirms it is significant and that there is interaction between age and grade.

Time-dependent effects

A Cox model was created with a time-dependent risk factor for age, $tt(\text{age})$, added. This helps model the interaction between age and time passed, as age has been shown not to be a constant risk factor, but dependent on time. The added risk factor $tt(\text{age})$ is a function which multiplies age by time. This corresponds to an age-effect that varies linearly. Results are given in table 8.

Table 8: Hazard ratios for Cox models with time-dependent age variable included

Risk factor	Hazard ratio	Confidence interval	p-value
Age ¹	1.056	0.97, 1.15	0.2
Grade	1.63	1.42, 1.86	<0.001
Time-dependent age variable	1.002	1.001, 1.004	<0.001

Using this model, we can plot the hazard ratio of age over time using the coefficients associated with the age and time-dependent age variables.

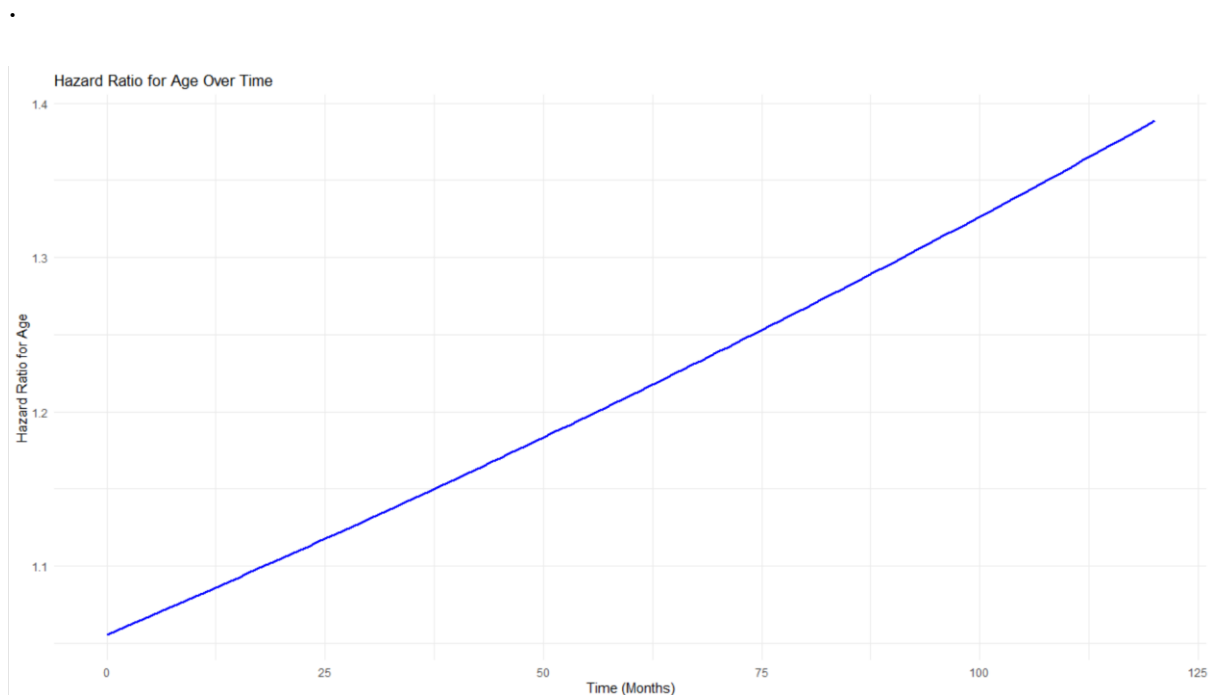


Figure 10: Time-dependent hazard ratio for age

This shows again the increasing nature of the hazard ratio associated with age over time. However, by the design of the age*time function, it still constrains the representation of the effect of age to be linear.

4.5 Non-linear Age Effects

Another Cox model was created with age modelled using cubic splines. Cubic splines, as described in the section 3.3, can be used to model non-linear associations and time-dependent hazards. The more flexible spline modelling can allow us to detect a possible non-linear relationship between the hazard ratio of age and time.

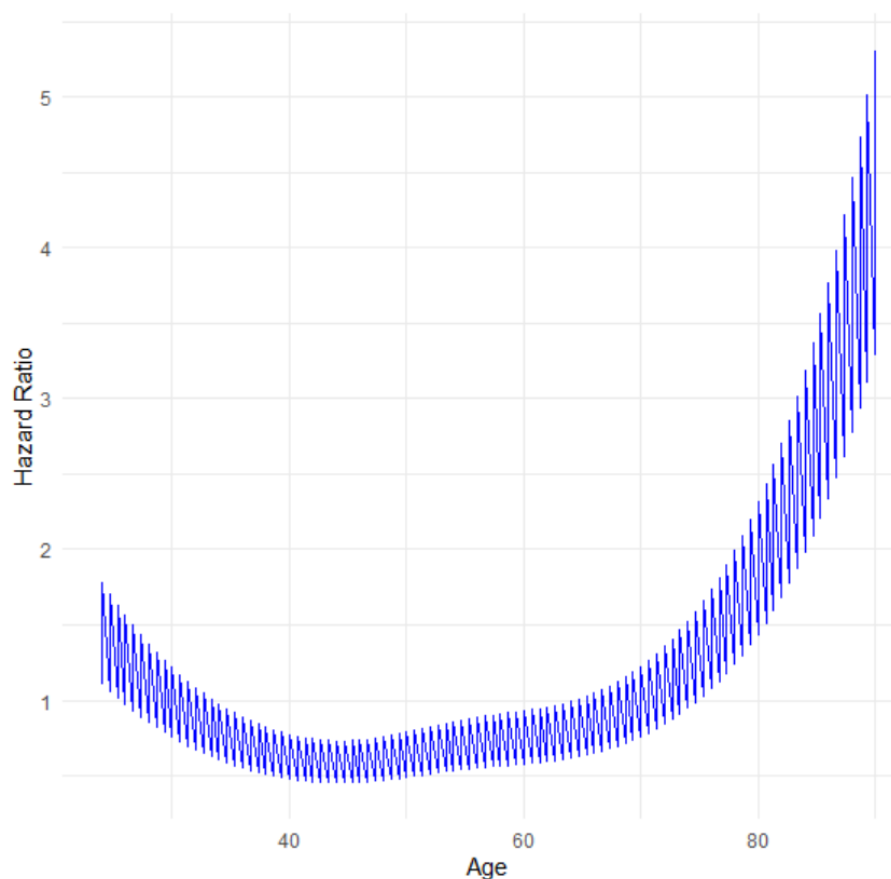


Figure 11: Spline estimate of the hazard ratio of age.

Above is the estimated hazard ratio for age when it is modelled using cubic splines. This tells a slightly different story than the linear interpretation where the hazard ratio for age is always increasing. Here the hazard ratio appears to be decreasing in early age, reaching a trough in the 40-50 range, before increasing, with this increase accelerating around age 70.

Another method to test the linearity of the effect of age is using multiple fractional polynomials [see section 3.4]. When modelling age using a fractional polynomial, the following output is produced.

Here, age is represented with powers, -0.5 and 3, and associated coefficients 1.9441 and 4.2735 and it's stated in the output that a transformation of the age value dividing it by 100 is applied such that

$$\text{age transformed} = I\left(\left(\frac{\text{age}}{100}\right)^{-0.5}\right) + I\left(\left(\frac{\text{age}}{100}\right)^3\right)$$

The hazard ratio of age therefore can be expressed as

$$HR(t) = \exp\left(\beta_1 \cdot \left(\frac{\text{age}}{100}\right)^{-0.5} + \beta_2 \cdot \left(\frac{\text{age}}{100}\right)^3\right)$$

Where: $\beta_1 = 1.9441$ and $\beta_2 = 4.2735$

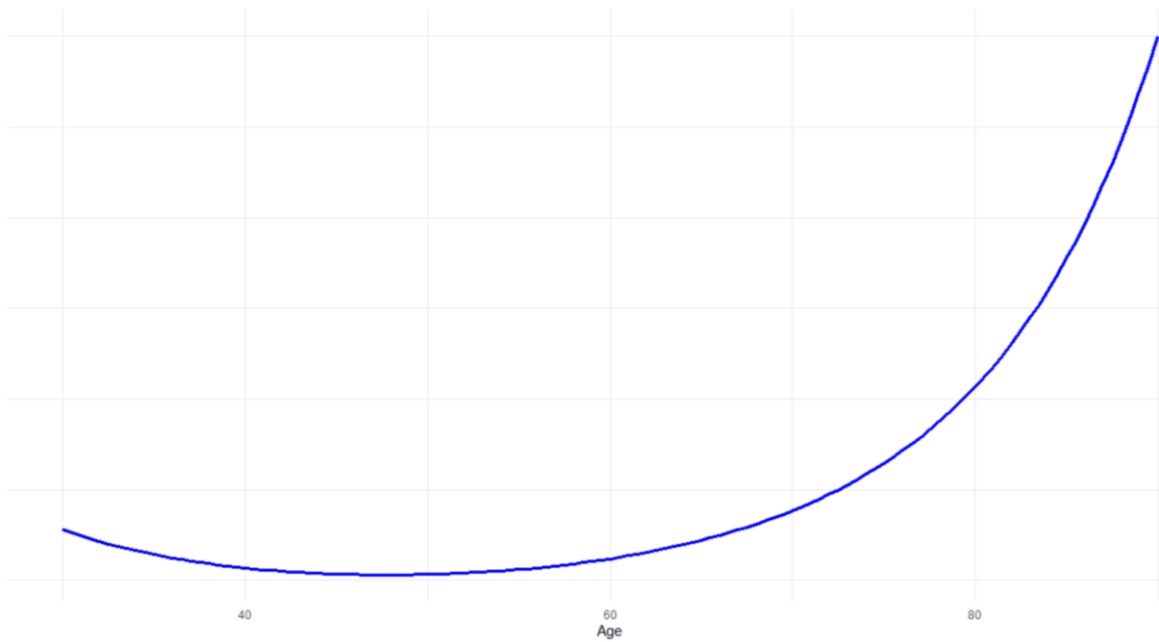


Figure 12: Fractional polynomial hazard ratio of age

This reaffirms the spline plot of the effect of age and confirms the previous readings that the hazard ratio of the age variable decreases from youth up to middle age before increasing in an accelerating manner.

Interaction effects with non-linear age

We have established that the association between age and hazard is non-linear. The effect of age when modelled linearly has also been shown to have time-dependence. Age does naturally increase with time and so is always a time-dependent variable in a literal sense. However, if the impact of age is linear, this does not result in a time-varying effect in the Cox model, as the hazard ratio will still be constant, because the difference in age between two patients will always be the same, even if they're both increasing. However, when a non-linear effect is incorrectly modelled as a linear effect, this can cause an apparent violation of the proportional hazards assumption, as the hazard ratio would no longer be constant. This explains how the findings of age to be non-linear, and its linear representation to be time-dependent are portraying the same information.

Given that temporal recalibration requires that effects are not time-dependent we have included age as a non-linear factor. As a result, we do not need to model time-dependent effects.

The other interaction effect previously discovered was between age and grade. This interaction is also dealt with by modelling the effect of age non-linearly. A Cox model was formed which included grade multiplied by each of the fractional polynomial terms for age, $I\left(\left(\frac{\text{age}}{100}\right)^{-0.5}\right)$ and $I\left(\left(\frac{\text{age}}{100}\right)^3\right)$. In this model, grade no longer showed a statistically significant interaction with age, as p-values of 0.87 and 0.19 were found for its interaction term with each age variable. This suggests that the original interaction discovered between age and grade was capturing non-linear effects of age, rather than grade altering the effect of age.

4.6 Temporal Recalibration

In this section we will consider the temporal recalibration of the risk prediction model that chooses from all the risk factors.

The full cohort model was developed on the full data set using multiple fractional polynomials, which chose two polynomial terms for age, included grade, progesterone receptors, tumour size, menopausal status and number of positive lymph nodes as linear terms and excluded hormonal treatment, chemotherapy and oestrogen receptors as they were not deemed significant variables in predicting survival. The β coefficients estimated by this model are presented in table 9.

Table 9: Coefficients associated with each risk factor in the full cohort model.

Variable	β	Standard error
Age^3	2.86	0.33
$Age^3 \times \ln(Age)$	13.64	2.32
Size (20 - 50mm) ²	0.42	0.07
Size (> 50mm)	0.75	0.09
Grade	0.30	0.07
Nodes	0.07	0.005
Menopausal status	0.36	0.11
Progesterone receptors	-0.0004	0.0001

Age is divided above by 100 when applying the fractional polynomial.

A temporally recalibrated model was created using the prognostic index from the full model as a sole risk factor on the data from the second period. The assumption when using the prognostic index from the full model as a predictor in the second model is that the effects of the risk factors do not change across the two periods. When fitting a model on the full data set using its prognostic index as a risk factor, its associated coefficient is one. This result is trivial as the prognostic index is designed to already fit the data perfectly. When using the same risk factor on the temporally recalibrated model on just the latter period, a coefficient of 1.07 with a standard error of roughly 0.05. As one is within two standard errors of the coefficient produced (0.96, 1.17), the assumption that

the effect of the risk factors doesn't change from the full data set to the latter period is reasonable. Finally, the period analysis model is developed using all the risk factors on just the latter period.

The cumulative incidence curves of all three models were plotted. The cumulative incidence curve is one minus the survival curve, i.e. the probability of not surviving over time.

Figure 13 displays cumulative incidence curves of the full cohort, temporally recalibrated and period analysis models.

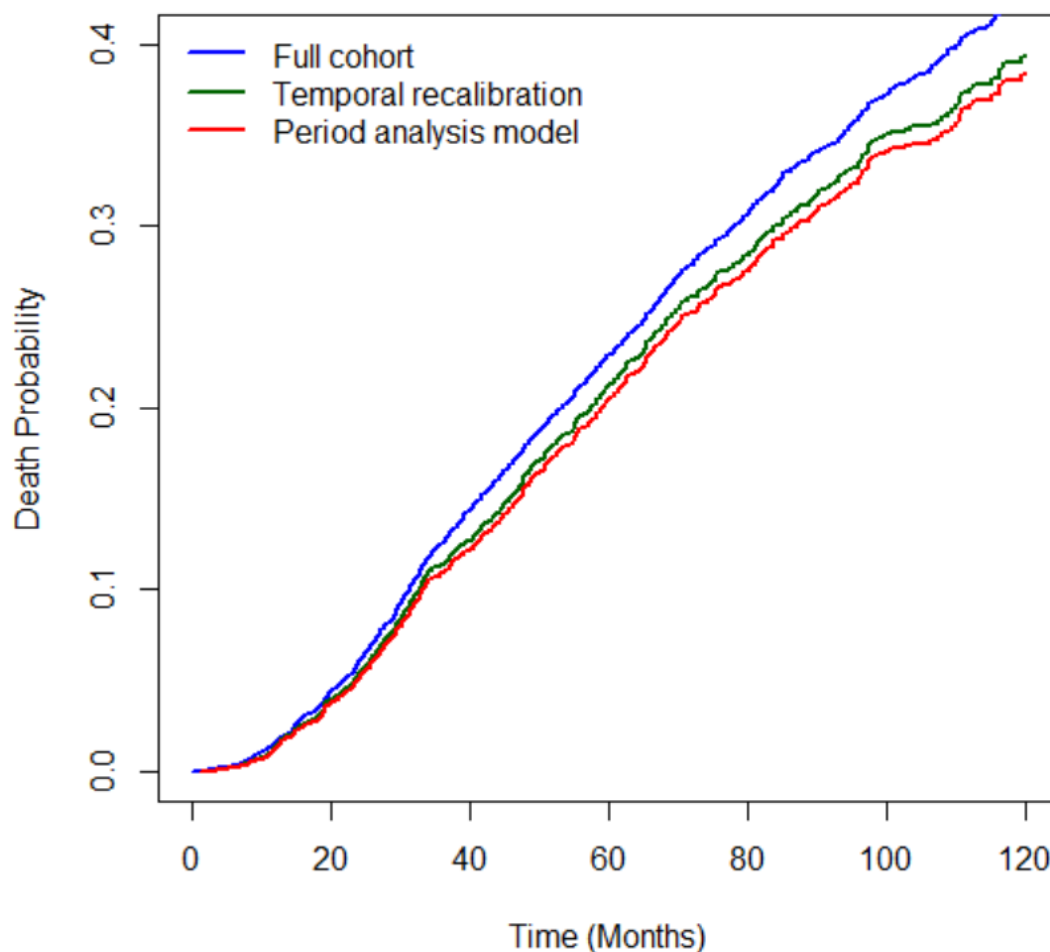


Figure 13: Cumulative incidence curves for the full cohort, temporal recalibration and period analysis models

The period analysis and temporally recalibrated models show very similar curves. This shows how the temporally recalibrated model reflects the improved survival outcomes in the latter period, while retaining more reliable risk factors estimated using the larger sample size of the full data set.

5. Discussion and Conclusion

In chapter 4, it was investigated and confirmed that survival probabilities were different in the two periods. The survival curves were visually and statistically distinct, and including period as a risk factor in the Cox model provided meaningful information about survival. The hazard ratios of grade and age were analysed and tested for proportionality and time-dependence. The time-dependence of age was used to create a linear increasing hazard ratio for age, before non-linear representations were found to best model the hazard ratio of age using splines or fractional polynomials. A temporally recalibrated model was created, and its survival curve was compared to that of the full cohort model and the period analysis model. Temporally recalibrating the full model was successful in replicating the significant improvements in the second period compared to the first. In testing whether the effect of the risk factors is comparable in the period analysis subset of the data as in the full data set, a coefficient near one was produced for the prognostic index from the full model in a separate model on just the latter period. This shows the validity of using temporal recalibration here, as the effect of the risk factors can be reasonably believed to be equivalent across the two data sets. This is somewhat contradictory to findings that the effect of grade is different across the two periods. However, when this was investigated initially only grade and age were included in models. Tumour grade could be presumed to exhibit collinearity with risk factors like tumour size, number of nodes and progesterone and oestrogen receptors, and many variables like tumour size and hormone treatment show different distributions across the two periods. In using full set of risk factors, some of the impact of grade can be absorbed by other correlated risk factors and shifting distributions could explain apparent shifting in effects of risk factors. This can help to explain why despite those early findings, the overall structure of the full prognostic index can maintain a consistent predictive performance across the periods. As such, temporal recalibration performs well here and achieves the

result of updating survival probabilities to be in line with more recent improvements. Period analysis achieves this same task, but its risk factor coefficients may be less reliable due to being calculated on a smaller sample size. In this example, a sample size of 1815 in the latter period may be sufficient, depending on how accurate the risk factor coefficients and hazard ratios are needed to be. An extension of the model presented here would model variation in the baseline hazard using non-linear methods such as cubic splines and fractional polynomial, as described in chapter 3.

The Rotterdam breast cancer data set used here is a very useful tool for survival analysis, with a large cohort of patients, long-term follow up and several detailed variables discussed in this report. This report focuses on time-to-death as the event outcome, but the data set also provides information on cancer recurrence and time to relapse. As such the Rotterdam study could be analysed using competing risks analysis. Competing risks can be events such as recurrence, cancer-specific death and other causes of death that are in competition and that can prevent or affect the probability of one another from happening. . In 2023, the temporal recalibration methods used here were extended to modelling with competing risks by Booth et al [6]. Competing risks can be events such as recurrence, cancer-specific death and other causes of death that are in competition and that can prevent or affect the probability of one another from happening. A further analysis would consider a temporal recalibration of the R study using the methods developed by Booth et al.

Disease progression can more suitably be conceptualised as a Markov process, where patients can transition between states such as diagnosis, cancer-free, recurrence and death. A logical next step would be the extension of temporal recalibration to more complex modelling frameworks including state transitions. To the best of my knowledge, temporal recalibration has not previously been applied to Markov models.

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